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Gemma 4 12B W4A16: evidence pack for a production approval decision

Date: 2026-08-15 **Incumbent:** InternVL3.5-8B (BF16, vLLM 0.19.0) — currently in production **Candidate:** Gemma 4 12B Unified QAT W4A16 (vLLM 0.25.1) **Question:** should the candidate be approved to run on the production estate?

Summary

Two benchmarks were run. **They disagree, and the disagreement is the most important thing in this pack.**

benchmark	ground truth	winner	margin
ICDAR SROIE 2019 (347 scanned receipts, 4 fields)	external, published	InternVL3.5	5.9 pts
Production corpus (165 documents, 14 fields)	generated with the images	Gemma 4	9.7 pts

The recommendation rests on the second, and on a hardware constraint that no amount of tuning can close. But the first is stated first, and in full, because it is the more credible instrument — we did not write its answer key.

1. What was measured, and how the comparison was controlled

Both models ran the **same prompt, parser, scorer, dataset split and batching configuration**. On the SROIE benchmark the only difference between the two runs was one line of configuration — `bootstrap.model.type`. On the production corpus the receipt prompt is shared repo-wide, so neither model has a prompt tuned to it.

This matters because most apparent model differences in this project turned out to be measurement defects, not capability. Five separate scoring bugs were found and fixed during this work; each was scoring a correct extraction as wrong. Full detail in §7.

2. Result A — public benchmark: InternVL3.5 wins

ICDAR 2019 SROIE Task 3, 347-image test split. Scanned Malaysian retail receipts. Four fields. Scored under the official exact-match protocol (**strict**) and a semantic policy (**lenient**).

Both models were measured on the **same sandbox hardware, 2 GPUs, data-parallel**, so this is a like-for-like comparison. Every figure below is read directly from the run artefacts, each of which records its own `model` and `total_images`:

model	artefact	images
internvl3-vllm	output_sroie_internvl3/sroie_summary.json	347
gemma4-12b-unified-w4a16-vllm	output_sroie_gemma4/sroie_summary.json	347

Each is accompanied by a `sroie_per_image.csv` of 347 rows, so any figure here can be traced to the individual receipts behind it.

	InternVL3.5-8B	Gemma 4 12B W4A16
strict overall	0.7889	0.7319
lenient overall	0.8892	0.8299
images	347	347

Per field:

field	InternVL3.5 strict	Gemma 4 strict	InternVL3.5 lenient	Gemma 4 lenient	favours
company	0.8559	0.8242	0.8934	0.8386	InternVL
date	0.9308	0.8444	0.9337	0.8501	InternVL
address	0.5879	0.4611	0.7637	0.6542	InternVL

field	InternVL3.5 strict	Gemma 4 strict	InternVL3.5 lenient	Gemma 4 lenient	favours
total	0.7781	0.7896	0.9625	0.9683	Gemma

The incumbent's result was independently reproduced on the production estate at strict 0.7882 / lenient 0.8885 — a difference of one field-document in 1,388 (\$6). The candidate could not be run there; see §8.

Two qualifications, both of which cut against over-reading this result:

- **Only date separates on non-overlapping confidence intervals.** The overall gap is credible because it is consistent in direction across three of four fields, not because any single field is individually significant.
- **Gemma 4 wins total** — the monetary field, and the one closest to what our production task actually needs.

InternVL3.5 also has a working resolution lever here and Gemma 4 does not: raising `min_tiles` from 1 to 6 bought +0.060 on address for 22% throughput. Gemma 4's equivalent knob is exhausted at its current setting — doubling it collapsed the model (on a 24-image subset, overall 0.8365 → 0.2752, address to 0.0000), reproduced on two independent corpora.

Read plainly: on unfamiliar scanned receipts, the incumbent is the better reader, and it has headroom the candidate lacks.

3. Result B — production corpus: Gemma 4 wins every field

165 synthetic business documents (bank statements, invoices, receipts), 14-field schema, answer key generated by the same pass that renders the images.

	clean (mixed types)	degraded (receipts)
Gemma 4 12B W4A16	0.957	0.930
InternVL3.5-8B	0.860	0.871

Per field on the degraded set — identical documents, identical answer key, identical scorer:

field	Gemma 4	InternVL3.5
LINE_ITEM_PRICES	0.5865	0.2762
LINE_ITEM_TOTAL_PRICES	0.8348	0.7877
IS_GST_INCLUDED	0.8485	0.7455
GST_AMOUNT	0.8727	0.8606
BUSINESS_ABN	0.9273	0.8485

field	Gemma 4	InternVL3.5
LINE_ITEM_DESCRIPTIONS	0.9993	0.8898
PAYER_NAME	1.0000	0.9212
BUSINESS_ADDRESS	0.9663	0.9493
LINE_ITEM_QUANTITIES	0.9927	0.9521
INVOICE_DATE	0.9939	0.9758
SUPPLIER_NAME	0.9989	0.9916
TOTAL_AMOUNT	1.0000	0.9939
DOCUMENT_TYPE, PAYER_ADDRESS	1.0000	1.0000

Gemma 4 wins or ties all fourteen. Two honest deductions from that:

- **IS_GST_INCLUDED measures nothing.** In this corpus it is fully determined by document type — every invoice false, every receipt true. On the receipts-only degraded set it is the constant true for all 165 documents, so a model scores it by answering the same way every time. It is worth 1/14 of the headline. Discount it.
- **LINE_ITEM_PRICES, the largest gap, is instruction-following, not reading.** The prompt asks for a unit price only where a line's quantity is 1. Gemma 4 applies that condition per line; InternVL3.5 writes NOT_FOUND at every position on all 55 receipts, and its score is flat across all degradation tiers precisely because its answer never varies. This is a real and operationally relevant difference, but it is about following a compound instruction, not about resolving pixels.

Excluding both, Gemma 4 still leads on every remaining field.

Robustness under image degradation

55 receipts × 4 quality tiers (the same 55 base documents at each tier, verified by filename):

tier	InternVL3.5	Gemma 4
clean	0.913	0.975
light	0.914	0.971
moderate	0.884	0.946
heavy	0.814	0.874

Neither model is more robust than the other. Both retain ~89% of clean accuracy at the heaviest tier (InternVL 0.892, Gemma 0.896) and the gap is flat at ~0.06 across all four tiers. Degradation is not a differentiator; Gemma 4 is simply better at every quality level. Classification was unaffected for both — 165/165 receipts correctly typed at every tier.

4. Why the two benchmarks disagree

They are different tasks, and the difference is not subtle:

	SROIE	production corpus
documents	Malaysian retail receipts	Australian business documents
capture	photographed / scanned, variable quality	rendered, clean
schema	4 fields, all scalar	14 fields, 4 of them multi-row lists
difficulty	reading small print off poor scans	applying a compound schema consistently

SROIE rewards raw optical robustness on unfamiliar layouts. Our production task rewards schema compliance across long line-item tables. The models differ on exactly that axis: pooled across tiers, the ten scalar fields separate the two models by a mean of 3.3 points and the four list fields by 12.6.

A model can be better at one and worse at the other, and the one that governs this decision is ours. That is an argument, not a dismissal — SROIE remains the honest counter-evidence and should be presented alongside.

5. Deployability: the constraint that cannot be tuned away

This is the strongest argument, because it is arithmetic rather than measurement.

On a 24 GiB card (L4 / A10G class), per data-parallel replica:

	InternVL3.5-8B (BF16)	Gemma 4 12B (W4A16)
weights	15.9 GiB	~9.6 GiB
KV cache remaining	2.27 GiB	~3× more
worst-case concurrency at max_model_len:	1.01×	3.43×
16384		

Data parallelism does **not** pool memory — each GPU runs a complete engine replica, so every replica faces the same ceiling.

The constraint is workload-specific, and the distinction matters. That concurrency figure is vLLM's worst case: $KV\ tokens \div max_model_len$, assuming every sequence uses the full context. Actual KV consumption scales with actual sequence length, so what batching buys depends entirely on how long the sequences are:

workload	per request	InternVL3.5 batching result
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workload	per request	InternVL3.5 batching result
SROIE receipt	2–3 tiles + thumbnail, 256 output tokens	8 concurrent fit; 3.3× measured
production extraction	18 tiles + thumbnail $\approx$ 4.9k vision tokens, up to 4096 output	no headroom; gain measured at zero

So InternVL3.5 batches perfectly well on light work — the production SROIE run in §6 used `max_num_seqs: 8` and reached 173.67 img/min across 4 GPUs. What it cannot do is batch **the production extraction workload**, where sequences genuinely approach the full 16,384-token context and eight concurrent 19-image prompts will not fit in a 23 GiB card at `gpu_memory_utilization: 0.90`.

That is the workload this decision is about, and it is where Gemma 4's roughly 6 GiB smaller weights convert directly into retained headroom: enabling batching on that path gains Gemma 4 approximately 9× (158.4 s against 1441 s) and InternVL3.5 nothing.

The constraint is a product of BF16 weights and an 18-tile image budget. The tile budget is a deliberate accuracy choice and is not available as a lever, so the only remaining levers for the incumbent are quantisation — which is precisely what the candidate already is — or more GPUs.

On like-for-like hardware and configuration (sandbox, 2 GPUs, data-parallel, batching enabled for both), Gemma 4 processed **107.2 img/min against 65.9** — 1.63×. Both figures are read from the run artefacts.

Consequence for capacity planning: on the extraction workload, the incumbent's throughput can only be increased by adding GPUs. The candidate's can be increased by using the ones we have.

6. Instrument validation

The SROIE benchmark was run on the production machine and on the development sandbox from the same commit:

	sandbox (DP=2)	production (DP=4)	Δ
strict	0.7889	0.7882	−0.0007
lenient	0.8892	0.8885	−0.0007

That delta is exactly **one flipped field-document out of 1,388** ($1 / (4 \times 347) = 0.00072$). Three individual field-documents changed and netted −1. company and address were identical to four decimal places on both machines.

The two runs used different data-parallel sharding (4 GPUs against 2), which changes batch composition and therefore the arithmetic order within a batch. Under greedy

decoding with *identical* batch composition the pipeline is bit-for-bit deterministic — confirmed by an accidental no-op re-run producing identical output on all four fields. So the observed movement is explained, not merely tolerated.

Why this matters for the decision: the harness — dataset loading, prompting, parsing, scoring — is identical code for both models. Validating it against production for the incumbent means the candidate's sandbox numbers were produced by an instrument known to agree with production to 0.0007.

Limit of that argument: it validates the harness, not the engine. Gemma 4 runs on vLLM 0.25.1 in a separate conda environment; that stack has not been exercised on the production estate. See §8.

7. Confounds and limitations

State these wherever the numbers are quoted. The comparison is controlled on data, ground truth, scorer, prompt and hardware, but not on these:

- **Engine version** — InternVL3.5 on vLLM 0.19.0, Gemma 4 on 0.25.1.
- **Precision** — BF16 against QAT W4A16. This is also the reason the candidate can be batched and the incumbent cannot, so it is not a confound that could be removed without removing the advantage.
- **Model generation** — Gemma 4 is roughly a year newer. Part of any margin is generational rather than architectural.
- **Image handling** — InternVL3.5 uses 448-px pre-tiling; Gemma 4 uses a soft token budget with pre-tiling off. Each is that model's correct configuration, but they are not the same mechanism.
- **Synthetic renders are not scans.** Production-corpus scores are an upper bound for real-world documents: BUSINESS_ADDRESS scores 0.95 there against 0.76 on scanned SROIE.
- **Five scoring defects were found and fixed during this work** — a text similarity fallback silently degraded to exact string matching, list fields compared as text rather than by type, quantity decoration (3x) compared as a string, descriptions missing from the register, and a cleaner that shifted index-aligned columns out of alignment. Each scored correct extractions as wrong. Gemma 4's clean-receipt accuracy moved 0.847 → 0.975 across those fixes and InternVL3.5's 0.752 → 0.913. **The ranking never changed, but the margin narrowed from ~10 points to ~6.** Treat any single-field difference as provisional until it has survived inspection of the raw output.
- **Re-scoring an unchanged run moved one model's mean by 0.049.** Any comparison must have both sides scored by the same code. Figures quoted from older logs are not comparable to figures produced today.

8. What this pack cannot answer

Honest gaps, all of which follow from the candidate not being approved to run on production:

1. **Gemma 4 has never run on the production estate.** All candidate figures come from the development sandbox. The harness is cross-validated (§6); the vLLM 0.25.1 engine and its environment are not.
2. **No production-hardware throughput figure for the candidate.** The 1.63× and the batching headroom are sandbox measurements and arithmetic respectively.
3. **No real-document accuracy for either model.** Both corpora are synthetic renders or public scans. Neither is our live document mix.
4. **address remains the weakest field for both models** (0.76 / 0.67 on SROIE) after every lever was exhausted. The residue is genuine small-print difficulty plus known answer-key errors.

9. Recommendation

Approve Gemma 4 12B W4A16 for a supervised production trial, on these grounds:

1. It is more accurate on the document types and schema we actually process, winning or tying all fourteen fields.
2. It is deployable in a way the incumbent is not **on the extraction workload**: at 18 tiles per document the incumbent has no batching headroom left on a 24 GiB card, while the candidate retains roughly 3.4×. This is a memory constraint, not a tuning gap — the only levers that would close it for the incumbent are quantisation, which is what the candidate already is, or more GPUs. (On light workloads such as SROIE the incumbent batches fine; the claim is specific to the documents we process.)
3. It is not more fragile — both models degrade identically under falling image quality, and classification is unaffected for both.

Not on the grounds that it is a better document reader in general. On public scanned receipts the incumbent is better by 5.9 points and retains a working resolution lever that the candidate lacks.

Proposed conditions

- **Consider approving a benchmark-only run as a first, smaller step.** Gaps (1) and (2) in §8 exist solely because the candidate may not execute on the production estate — so they cannot be closed before an approval, only by one. A time-boxed SROIE run is a much narrower request than a deployment: one configuration change, roughly four minutes of inference, read-only against a public dataset, no production data touched, and no change to the serving path. It would produce a

like-for-like production figure for the candidate against the incumbent result in \$2, which is the single largest improvement available to this evidence base.

- Keep the incumbent deployable for rollback throughout the trial.
- Re-measure on a sample of real documents before any unsupervised cutover; gap (3) is the one that most affects operational risk.